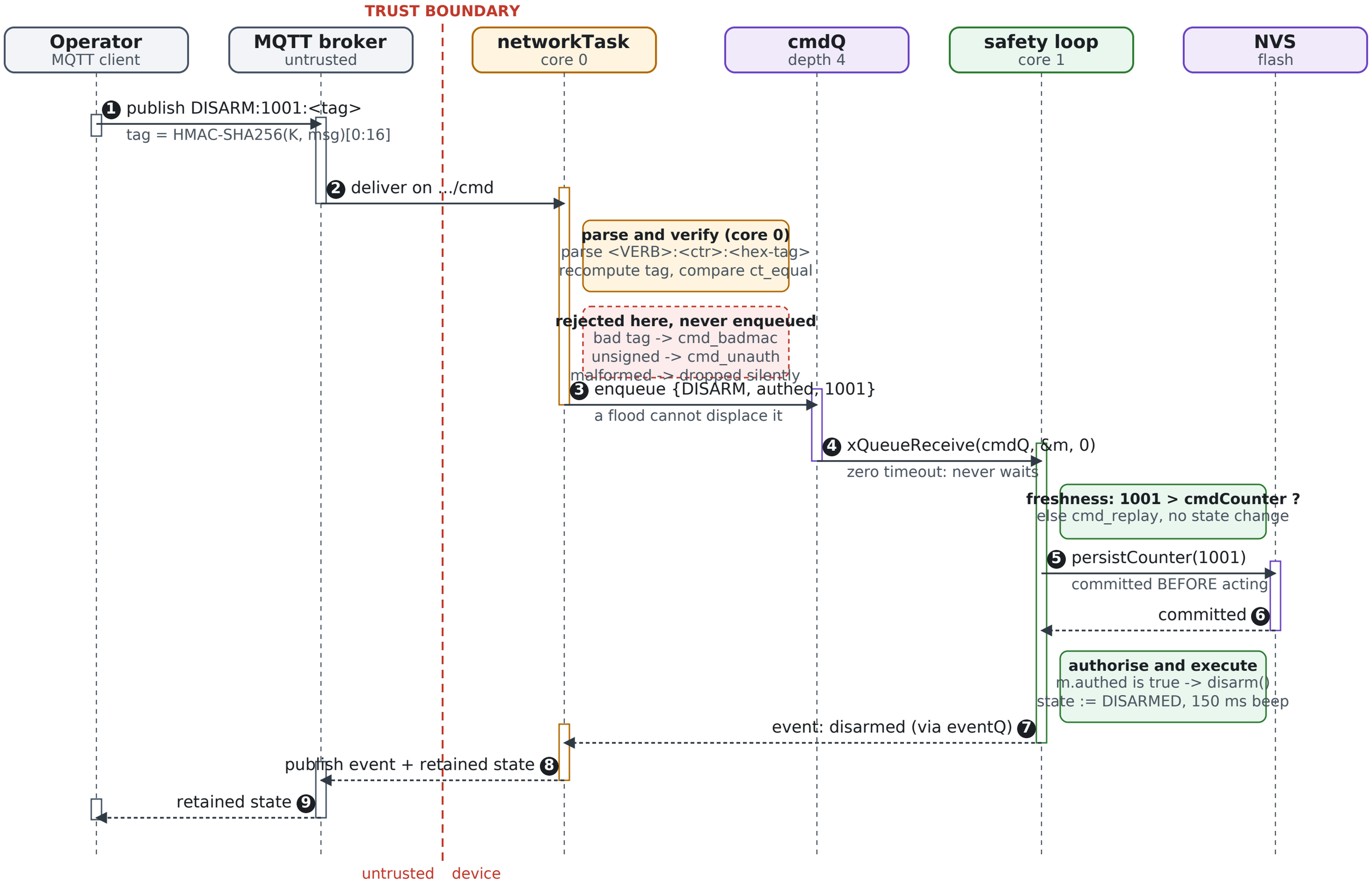
Smart Home Surveillance System v2 - Command Execution Sequence

Authenticated remote DISARM: HMAC-SHA256 over a reboot-persistent counter, verified on core 0



LEGEND

- Outside the device (untrusted)
- Core 0: network domain
- Core 1: safety domain
- control flow
- Internal shared resource
- Rejection path
- reporting, not control

Step 5 commits the counter to flash before the command is acted on, so an interruption can at worst burn a counter value: it cannot leave the command replayable. The operator gets no direct acknowledgement: there is no command-ack protocol, so the outcome is inferred from the retained state topic and the event stream. Steps 7 to 9 are therefore best-effort reporting at MQTT QoS 0 (at-most-once delivery).